How Much Did the Model Read? Truncation, Coverage Bias, and the Reliability of LLM-Derived Measures from 10-K Filings

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ABSTRACT

Studies applying large language models to annual-report text almost always truncate each filing to a fixed budget of words or tokens. We show that this routine step is not neutral for the questions such studies ask. Across 785 10-K filings whose size we take from the filer's own reported public float, a 2,000-word budget truncates 99.1% of filings and leaves the median filing showing under 9.5% of its narrative; narrative length rises with firm size (Spearman $\rho = 0.161$, $p =< 0.001$), so a fixed budget reads proportionally *less* of a larger firm's filing. How much of the document the model saw therefore becomes a function of firm size — the variable size-effect studies regress on. Even a 32k-token budget truncates 39.7% of filings.

We propose a window-level protocol that removes the confound: narrative sections are extracted complete and scored in fixed 1,200-word windows, so every filing is read in full and every judgment is made on an identically sized unit. Using it we measure *interpretive complexity* as disagreement between three analytical personas under a single fixed model, on 59 filings from 30 firms, and we pre-registered the analysis before scoring.

The measure does not support the cross-sectional inference it was built for, and we report why rather than only that. At the window level it is real: inter-persona disagreement runs 1.64× the same-persona noise floor and exceeds a persona-free baseline. But filing-level split-half reliability is 0.239 and only 78.3% of window-level variance lies between firms, so filing means are mostly noise. The size gradient is accordingly null ($p = 0.235$), and a pre-registered subsampling test shows that unequal window counts alone manufacture a gradient larger than the one observed. We report the diagnostics any such measure should be required to pass.

KEYWORDS

large language models, financial disclosure, textual analysis, measurement reliability, 10-K filings

1 INTRODUCTION

A growing literature applies language models to the narrative sections of annual reports. Nearly all of it faces a mundane engineering decision first: filings are long, budgets are finite, so the text is truncated. The decision is usually reported in a sentence, if at all.

That decision is not innocuous for the questions the literature asks. Firms differ in how much they write, and they differ systematically. A *fixed* budget therefore reads a *variable* fraction of the document, and that fraction is correlated with firm size. Any cross-sectional comparison by size is then confounded with a coverage gradient the researcher introduced.

This paper makes three contributions, and we think the third is the most useful.

We document the confound at scale. Across 785 10-K filings, narrative length rises with public float ($\rho = 0.161$, $p =< 0.001$), and a 2,000-word budget truncates 99.1% of them. Figure 1 gives the fraction read under budgets in common use.

We propose a protocol that removes it, and show what it costs to build: reliable complete-section extraction from EDGAR HTML is substantially harder than it looks, and the failures are silent under a word cap.

We report an honest negative result with the diagnostics that explain it. Our measure of interpretive complexity does not separate size tiers. Rather than leave that as a null, we show that the measure is real at the window level but unreliable once aggregated to the filing, and we quantify how much apparent gradient the aggregation itself can manufacture. We believe measures of this class should be required to report these numbers before they are used for inference.

2 RELATED WORK

Textual complexity in disclosure. The accounting literature has long measured readability and complexity of annual reports and related them to firm characteristics and market outcomes [1, 5, 7, 8]. This work establishes both that narrative length varies systematically across firms and that naive readability proxies behave poorly on financial text [8].

Disagreement as a quantity. Treating divergence of opinion as economically meaningful has a long history [2, 9]. Our measure is a mechanical analogue: rather than dispersion among human analysts, we measure dispersion among analytical lenses applied to the same text by one model.

LLMs on financial text. Recent work applies language models directly to disclosure [4, 6], and a parallel strand warns about look-ahead and identity leakage when models have memorised the firms or periods in question [3]. We restrict the sample to filings after a conservative training-cutoff bound. **[TODO: verify every citation in refs.bib]**

3 DATA AND PROTOCOL

3.1 Sample

30 firms, stratified into three size tiers, contributing 59 filings: the most recent 10-K and 10-Q per firm filed after the cutoff. The sample is frozen in a manifest pinned by (ticker, form, date), so it cannot drift as firms file again, and the analysis reproduces offline from cached filings.

3.2 Complete-section extraction

We extract Management's Discussion and Analysis (MD&A) and Risk Factors as complete sections, with no word cap anywhere. Doing so reliably is harder than it sounds, and the difficulty is worth reporting because a word cap hides it.

Two failure classes appeared in our corpus. First, *boundary overrun*: a section span running past the end of the section. One 10-K's "MD&A" initially measured 107,257 words because the span swallowed the financial statements; under a 2,000-word cap this is invisible. Second, *header deletion*: some filers render each item heading inside its own single-cell table, and a sparse-table filter — a standard HTML-cleaning step — removed them entirely, so the section had no recoverable start. That defect cost one large-cap filing its whole Risk Factors section and silently truncated three large-cap MD&As.

Both were fixed before scoring. After the fix, MD&A ranges from 2,675 to 30,340 words, all 59 filings yield a usable MD&A, and 49 yield Risk Factors; the 10 that do not are 10-Qs incorporating the 10-K's risk factors by reference, verified individually.

The direction of these errors matters: they truncated *large-cap* filings, which is the same direction as the confound the protocol exists to remove.

3.3 Window-level scoring

Each section is split into fixed 1,200-word windows. Three analytical personas — value, sentiment and fundamentals — score every window independently under one model (openai/gpt-oss-20b), with no interaction between calls. This is a prompt ensemble, not a multi-agent system: the model is held fixed and only the analytical lens varies. The run comprises 667 windows and 2,001 persona calls, of which 363 windows are MD&A.

All calls were served by a single backend at `bf16` precision with fallbacks disabled; the pipeline records which backend served each call and fails the run if more than one did. Routing across backends at different quantisations would make part of the measured disagreement a quantisation artifact.

3.4 Measures and pre-registration

Window-level disagreement is the mean pairwise cosine distance between the three personas' reasoning embeddings (primary) and the standard deviation of their scores (secondary). Filing-level disagreement is the unweighted mean over that filing's windows; firm-level is the mean over its filings. The firm is the unit of analysis for every confirmatory test.

The analysis plan was committed before any filing was scored (`fec1eb2`, amended 2 times before scoring, final `c22f5dd`). It fixed the primary outcome (MD&A only), the single confirmatory test, the window-count battery and its pass criteria, the rule for retiring the secondary measure, and what counts as a null. Deviations are logged and reported.

4 THE COVERAGE CONFOUND

In our sample the retired cap truncated 59 of 59 filings. Median complete narrative length is 15,488 words for large-caps and 7,285 for small-caps, and the median share read runs the other way: 12.9% against 27.5%. Spearman's ρ between size rank and share read is -0.415 ($p = 0.001$).

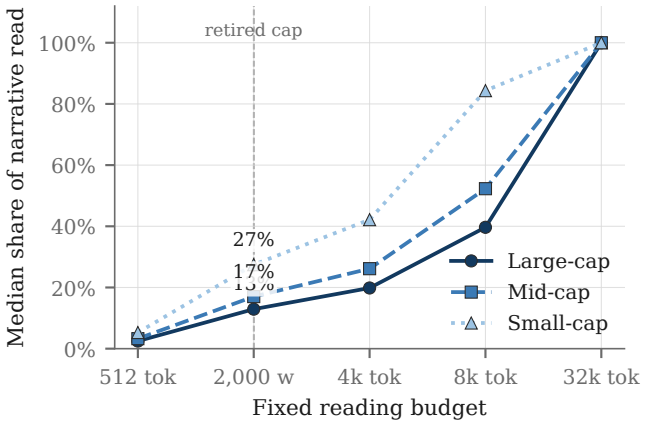


Figure 1: Median share of a filing's narrative actually read under budgets common in the literature, by size tier, in our 59-filing sample. Token budgets converted at 0.75 words per token. At the 2,000-word cap this project retired, the median large-cap filing was read at 12.9% against 27.5% for small-caps.

Table 1: Coverage at scale: 785 10-K filings across public-float deciles. "Truncated" is the share of filings whose narrative exceeds the budget. Smallest and largest refer to float deciles.

Budget	Median read	Smallest	Largest	Truncated
512 tokens	1.8%	2.6%	1.8%	100.0%
2,000 words	9.5%	13.4%	9.3%	99.1%
4k tokens	14.6%	20.5%	14.3%	97.7%
8k tokens	29.3%	41.1%	28.5%	93.8%
32k tokens	100%	100%	100%	39.7%

At scale. To test whether this is a property of our sample or of the filing population, we repeated the length measurement on 785 10-K filings drawn across public-float deciles, taking size from `dei:EntityPublicFloat` as reported by the filer on its own cover page — an SEC-filed value, not a reconstruction from market data. Extraction succeeded on 98.1% of attempts. Table 1 gives the result.

Narrative length rises with public float ($\rho = 0.161$, $p =< 0.001$, $n = 785$; elasticity of log length on log float 0.0527). The coverage gradient is weaker here than in our stratified pilot — a 1.4× smallest-to-largest ratio against 2.1× — which is expected, since the pilot contrasts extremes and the population is a continuum. We report both. The blunter fact needs no gradient at all: under budgets in routine use, nearly every 10-K is truncated, and the median filing is read at under 9.5% at 2,000 words.

5 DOES TRUNCATION CHANGE THE ANSWER?

Using the window-level output we recomputed the tier gradient as it would have been measured had the model read only the first

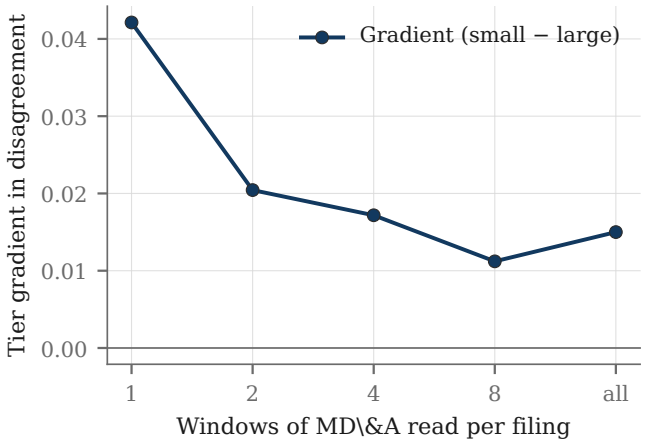


Figure 2: Tier gradient in the primary measure as a function of how many windows of MD&A the model is allowed to read, holding the model, the filings and the scored text fixed. Shaded band is a 95% bootstrap CI.

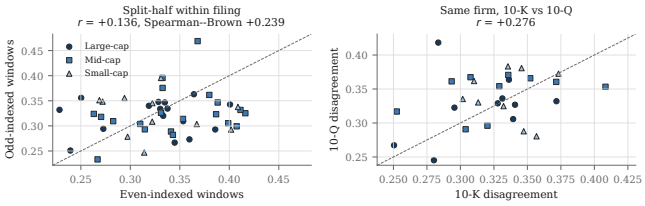


Figure 3: Reliability of the filing-level measure. Left: odd- against even-indexed windows within the same filing. Right: the same firm's 10-K against its 10-Q. Both sit far from the identity line.

k windows of each filing. The gradient is largest under the harshest truncation (+0.0421 at k=1) and shrinks as coverage grows (+0.0150 complete); for the secondary measure the sign reverses entirely (+0.1397 at k=1 against −0.0544 complete). Truncation therefore *inflates* the apparent size effect, in the direction the coverage analysis predicts.

We state the limit plainly: at $n = 30$ firms every one of these intervals spans zero, so this is directionally consistent evidence, not a demonstration. It is also a simulation of truncation using windows scored in isolation, which is not identical to a model reading a truncated document.

6 IS PERSONA DISAGREEMENT A MEASURE?

It is real at the window level. Running the same persona three times on the same window at the production temperature gives a noise floor; inter-persona distance runs 1.64× that floor, and exceeds a persona-free baseline (three copies of one neutral analyst prompt) by 0.1395. The analytical lens does measurable work. *Caveat:* this subset was interrupted by provider rate limiting and covers 27 large-cap windows, so it is indicative only.

It is discriminant. Filing-level disagreement is uncorrelated with length ($\rho = -0.148$), Fog ($\rho = -0.007$), and Loughran–McDonald

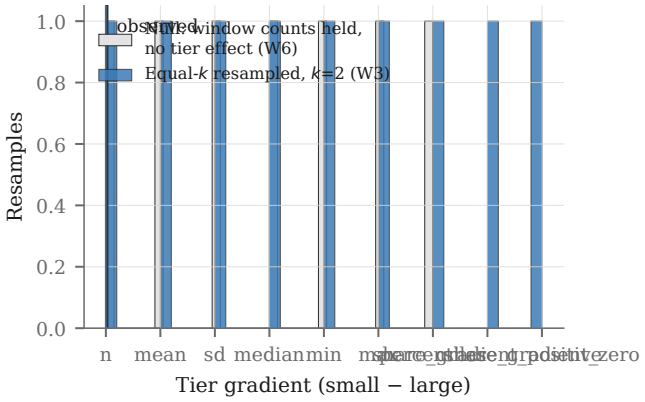


Figure 4: The observed gradient against two reference distributions: equal-k resampling, which removes the window-count difference, and a null in which real window counts are held fixed but all tier signal is destroyed.

negative, uncertainty and litigious shares (all $|\rho| \leq 0.09$), none significant. It is not length, readability or tone under another name.

It is encoder-robust. Recomputing the whole measure with `all-mpnet-base-v2` and `bge-small-en-v1.5` gives window-level rank correlations of +0.755 and +0.713 against the primary encoder, and all three agree there is no significant tier effect ($p = 0.18$ to 0.38).

But it does not survive aggregation. Split-half reliability within a filing — odd against even windows — is $r = 0.136$, Spearman–Brown corrected 0.239. Test–retest between a firm's 10-K and 10-Q is $r = 0.276$ ($p = 0.147$). A variance decomposition agrees: 78.3% of window-level variance is within filings, and the firm-level intraclass correlation is 0.077. Filing means over roughly six windows are therefore mostly noise, and firm means are worse.

This is the paper's central measurement finding. A measure can be real at the unit where it is computed and still be unusable at the unit where the hypothesis lives.

7 INTERPRETIVE COMPLEXITY AND FIRM SIZE

We report the secondary numeric measure first, as pre-registered. Firm-level means are 0.5978 (large), 0.5981 (mid) and 0.5434 (small): non-monotone, with small-cap *lowest*, against a one-sided p of 0.94. The pre-registered retirement rule did not trigger ($\rho = 0.046$ against complete MD&A length, $p = 0.727$, $n = 59$), so the measure is flat but is not a length measure in disguise.

The primary measure gives firm-level means of 0.3202 (large), 0.3355 (mid) and 0.3352 (small). The gradient is 0.0150 in the predicted direction but not monotone. The confirmatory test is not significant: $J = 169.0$, $z = 0.722$, one-sided $p = 0.235$, with a label-permutation p of 0.242.

7.1 Is the gradient an averaging artifact?

Window count scales with filing length and so with tier ($\rho = 0.372$, $p = 0.004$), so a filing mean averages far more windows for a large firm than a small one. Averaging more draws reduces the variance

of a mean without shifting its expectation, but composition and window-size effects can bias it, so we tested this directly rather than argue about it.

Equalising window counts by subsampling $k = 2$ windows per filing, the primary sign holds in 0.885 of draws against a pre-declared threshold of 0.90: the criterion is not met. The window-level mixed model agrees, with a Small-vs-Large coefficient of 0.0142 [-0.0126, 0.0411] ($p = 0.299$).

The sharpest result is the null simulation. Holding each filing's real window count fixed and destroying all tier signal, unequal window counts alone produce a Small-minus-Large gradient of 0.0218 at the 95th percentile — *larger than the 0.0150 we observed*, which sits at the 91.4th percentile of that null ($p = 0.086$). A gradient of this size is within what the aggregation manufactures on its own.

8 WHAT THIS MEANS

Under the pre-registered definition the result is a null on three independent grounds: the confirmatory test is not significant, the tier means are not monotone, and the window-count gate fails. We report it as the headline.

The useful reading is not that smaller firms write no more ambiguously. It is that a filing-level measure built this way cannot answer the question. The construct shows signal where it is computed, loses it on aggregation, and the aggregation itself can produce a gradient of the size reported. Any study aggregating LLM judgments over document chunks faces the same arithmetic, and we would argue for reporting split-half reliability, an intraclass correlation, and an equal-k subsampling check as a matter of routine.

9 LIMITATIONS

30 firms is a measure-validation sample. Size tier stands in for a point-in-time size measure not yet available. One model family produces the main results. We report no market outcome. The persona prompts are a design choice, and a different set of lenses would give a different measure. The noise-floor subset was cut short by rate limiting and is large-cap only. Deviations from the pre-registered plan are logged in the repository and summarised in the appendix.

10 CONCLUSION

Truncating filings to a fixed budget makes coverage a function of firm size, and at scale nearly every 10-K is truncated under budgets in routine use. A window-level protocol removes that confound. Applying it, our measure of interpretive complexity is real at the window level, discriminant, and encoder-robust — and still too unreliable at the filing level to support the cross-sectional claim it was built for. We report that, and the diagnostics that establish it, as the more useful result.

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